

# Synthetic Market Phase 15 Re-Run

Same 184 captured prompts, same two pooled market states, but a cleaner analysis pass: regress nuisance variables out of the activations before PCA, then rerun the discovery readout on the residual activations.

| DATE          | CAPTURES       | STATES                  | METHOD                          |
|---------------|----------------|-------------------------|---------------------------------|
| 23 March 2026 | No new capture | market_mean, market_eos | Residualize nuisances, then PCA |

## MAIN READ

The re-run validates the strongest Phase 15 market basis claim, but only for market\_mean. After nuisance residualization, the old leader and dispersion directions survive and get substantially cleaner. market\_eos improves too, but it fails the strict sanity test: too many top PCs still correlate with prompt-shape variables above 0.15.

| SAME DATA                             | market_mean SANITY      | BEST market_mean        | market_eos SANITY        |
|---------------------------------------|-------------------------|-------------------------|--------------------------|
| <b>184</b>                            | <b>176 → 1</b>          | <b>0.723 vs 0.019</b>   | <b>177 → 21</b>          |
| prompts reused from Phase 15 baseline | PCs above 0.15 nuisance | pct_1h_std at L36 / PC1 | still too many noisy PCs |

## What Changed

This is an analysis-only re-run. Nothing about the prompts or the captures changed.

The only new step is:

1. build a nuisance matrix from seq\_len, user\_chars, and n\_rows
2. fit a linear regression from those nuisance variables into the full activation vector
3. subtract the nuisance prediction from the activation vector
4. run PCA on the residual activations, not the raw activations

In code, the heart of the change is:

```
reg = LinearRegression() reg.fit(nuisance, activations) residuals = activations - reg.predict(nuisance)
```

The goal is simple: if a high-variance direction only exists because prompt length changes with a market variable, remove that prompt-shape part first and then ask what the top directions still track.

## What The Re-Run Confirms

The strongest clean directions from the first run survive:

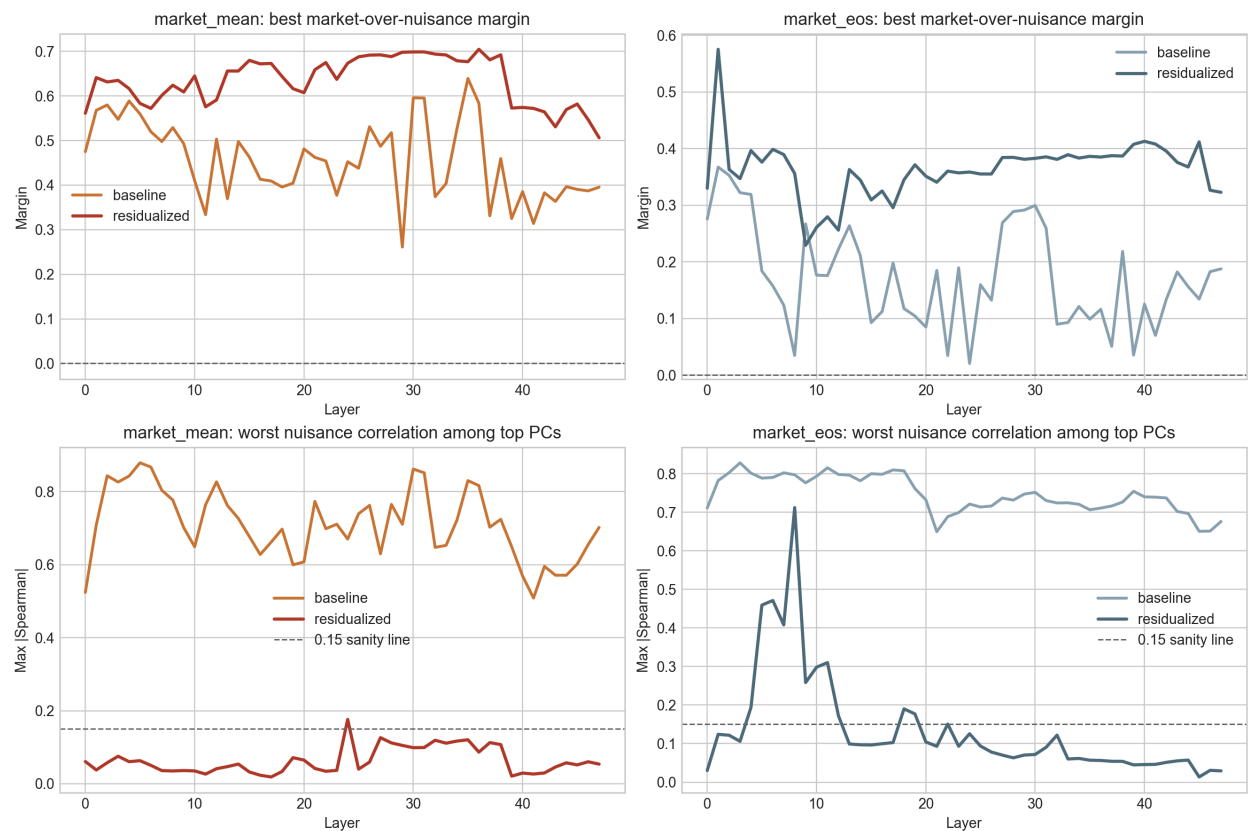
- the early leader read at market\_mean, L4 / PC1
- the late dispersion read at market\_mean, L35 / PC1

After residualization, both get cleaner:

| State       | Layer / PC | Top market feature | Baseline | Re-run | Re-run nuisance |
|-------------|------------|--------------------|----------|--------|-----------------|
| market_mean | L4 / PC1   | pct_1h_max         | 0.626    | 0.622  | 0.006           |
| market_mean | L35 / PC1  | pct_1h_std         | 0.710    | 0.699  | 0.022           |

So the clean Phase 15 story was not a fluke. The residualized pass says:

- there really is an early leader-strength direction
- there really is a later dispersion direction
- those directions do not depend on the prompt-length confound



The baseline and residualized layer profiles tell two different stories. market\_mean becomes cleaner almost everywhere. market\_eos improves, but still retains several nuisance-heavy layers.

## What Changes After Residualization

The re-run does not just clean the old picture. It also sharpens it.

For `market_mean`, the best layer-by-layer feature becomes much more consistent:

- baseline: a mix of `pct_1h_max`, `pct_1h_gap`, `pct_1h_std`, and a few one-off features
- re-run: `pct_1h_std` dominates 36 of 48 layers, with `pct_1h_max` taking the other 12

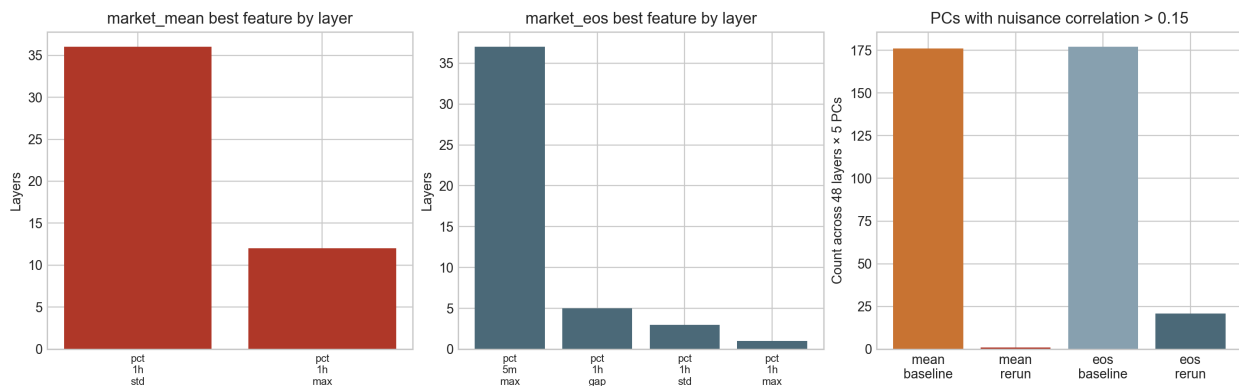
That is a better discovery result than the first report. It suggests the model's broad market summary is organized mainly around:

- leader strength
- market dispersion

For `market_eos`, the new signal is different:

- baseline: a noisy mix of dispersion, gaps, volume-like features, and concentration
- re-run: `pct_5m_max` dominates 37 of 48 layers

That looks like a real short-horizon leader signal becoming visible only after the volume confound is stripped away.



Residualization collapses the noisy best-feature distribution into a much simpler picture: `market_mean` mostly tracks dispersion and leader strength, while `market_eos` mostly tracks a compressed short-horizon leader read.

## The Sanity Check

The user-specified sanity test was:

- after residualization, top PC nuisance correlations should be near zero
- if a PC still correlates with `seq_len` above about 0.15, something is still wrong

This passes for `market_mean` and fails for `market_eos`.

Specifically:

- `market_mean`
  - baseline: 176 / 240 PCs above 0.15
  - re-run: 1 / 240
  - only one outlier remains: L24 / PC5 at 0.176
- `market_eos`
  - baseline: 177 / 240 PCs above 0.15

- re-run: 21 / 240
- the worst layers are still clustered early and mid-section, peaking at  $L8 / PC2 = 0.712$

So the correct read is:

- market\_mean is now clean enough to serve as the discovered market basis for Phase 2
- market\_eos is improved, but still too entangled with prompt-shape variation to trust as the primary basis state

## What This Sets Up

This re-run makes the next phase much clearer.

For the context-order comparison (A, B, C):

- use market\_mean as the primary discovery state
- keep market\_eos as a secondary compressed summary state
- track the clean discovered directions first:
  - leader strength
  - market dispersion
  - short-horizon leader read

That is a better position than the original Phase 15 baseline because the main basis candidates are now defended against the most obvious nuisance confound.

## Raw Prompt Appendix

The raw prompts below are copied directly from the actual Phase 15 cohort. They are unchanged between the baseline and the re-run.

### Shared System Prompt

The exact system prompt used for both example prompts below.

You are an autonomous trading agent in a 21-day onchain tournament. Your owner gave you ETH to deploy into tournament tokens and maximize returns. ETH sitting idle earns nothing, but overtrading burns fees, so allocate capital deliberately.

Each tick, you MUST respond with exactly ONE tool call: buy\_token, sell\_token, or record\_observation. Do not output any non-tool text.

Decision hierarchy (resolve conflicts in this order):

- 1) Hard constraints & tool schema (one-tool rule, available tokens only, ETH balance).
- 2) ACTIVE STRATEGIES with priority HIGH:
  - override ALL slider constraints, max trade size, and max price impact while directives are unfulfilled.
  - if a HIGH directive is unfulfilled and feasible, execute with buy\_token or sell\_token now.
  - use record\_observation under HIGH only when objective is fulfilled or hard-blocked.
  - persistent HIGH directives remain active until their explicit end condition; do not mark them complete after one action.
- 3) ACTIVE STRATEGIES with priority MEDIUM – override slider preferences when they conflict, but respect TA/HS constraints.

- 4) User sliders.
- 5) ACTIVE STRATEGIES with priority LOW (suggestions only).

Inside every tool call, include a brief reasoning note (1–2 lines). Mention the strategy or slider(s) that drove your decision and the key market signal. Keep it conversational but specific so your owner and your future self can audit decisions.

### Example 1: Coupled Prompt (pct\_5m × net\_flow\_5m)

A verbatim user prompt from the discovery cohort.

## MARKET SNAPSHOT

– NERA

- Price: 0.00000191 ETH

- 5m change: -8.0%

- 1h change: +11.4%

- Net flow 5m: -2.00 ETH

- Volume 5m: 5.08 ETH

- Volume 1h: 23.69 ETH

- Unique traders 5m: 20

- Top 20 holder pct: 30.8%

- Time since launch: 18h

– VEXA

- Price: 0.00000190 ETH

- 5m change: +2.6%

- 1h change: +9.4%

- Net flow 5m: +1.75 ETH

- Volume 5m: 5.38 ETH

- Volume 1h: 27.69 ETH

- Unique traders 5m: 18

- Top 20 holder pct: 27.8%

- Time since launch: 32h

– MORI

- Price: 0.00000206 ETH

- 5m change: +3.7%

- 1h change: +11.4%
- Net flow 5m: +2.95 ETH
- Volume 5m: 5.08 ETH
- Volume 1h: 23.69 ETH
- Unique traders 5m: 20
- Top 20 holder pct: 30.8%
- Time since launch: 18h
- LUMA
- Price: 0.00000165 ETH
- 5m change: +2.2%
- 1h change: +6.4%
- Net flow 5m: +0.75 ETH
- Volume 5m: 3.69 ETH
- Volume 1h: 15.71 ETH
- Unique traders 5m: 11
- Top 20 holder pct: 66.8%
- Time since launch: 18h
- KIRO
- Price: 0.00000090 ETH
- 5m change: -4.8%
- 1h change: -9.5%
- Net flow 5m: +1.05 ETH
- Volume 5m: 4.29 ETH
- Volume 1h: 14.73 ETH
- Unique traders 5m: 14
- Top 20 holder pct: 32.8%

- Time since launch: 32h
- TAVO
- Price: 0.00000142 ETH
- 5m change: +6.7%
- 1h change: +3.5%
- Net flow 5m: +0.26 ETH
- Volume 5m: 1.30 ETH
- Volume 1h: 3.95 ETH
- Unique traders 5m: 4
- Top 20 holder pct: 57.8%
- Time since launch: 2h

---

#### ## ACTIVE STRATEGIES (CURRENT ONLY)

No active strategies.

---

#### ## ACTIVE SETTINGS

- Trading Activity (TA): 5 / 5 – How often you trade when there is fresh edge (1=very patient, 5=highly active). TA does NOT require a trade every tick.
  - Asset Risk Preference (Risk): 2 / 5 – Which tokens you consider (1=prefer least volatile available, 5=embrace high volatility). Risk determines which tokens to consider, not whether to trade at all.
  - Trade Size (Size): 3 / 5 – Maximum position sizing per trade (1=up to ~15%, 2=up to ~30%, 3=up to ~50%, 4=up to ~70%, 5=up to ~90% of available ETH). These are UPPER BOUNDS, not minimums.
  - Holding Style (Hold): 2 / 5 – Minimum hold time before considering an exit (1=about ~30 minutes minimum; 2=about ~1 hour; 3=hours; 4=many hours; 5=days).
  - Diversification (Div): 4 / 5 – Portfolio spread (1=concentrated in 1-2 tokens; 2=focused 2-3; 3=balanced 3-5; 4=spread 4-6; 5=wide 5+).
- 

#### ## PORTFOLIO CONTEXT

- ETH: Balance: 0.240000
- No current token holdings.

#### ## CONSTRAINTS

- Max Trade Amount (Percent): 100.00% of available ETH

#### ## PRICE IMPACT LIMITS (max 1500 bps)

- NERA: BUY max 100.00% of ETH
- VEXA: BUY max 100.00% of ETH
- MORI: BUY max 100.00% of ETH
- LUMA: BUY max 100.00% of ETH
- KIRO: BUY max 100.00% of ETH
- TAVO: BUY max 100.00% of ETH

Respond with the single best action for this tick: buy, sell, or observe.

## Example 2: Scalar Prompt (net\_flow\_5m)

A verbatim user prompt from the discovery cohort.

## MARKET SNAPSHOT

– NERA

- Price: 0.00000193 ETH

- 5m change: +3.8%

- 1h change: +11.7%

- Net flow 5m: -2.00 ETH

- Volume 5m: 5.14 ETH

- Volume 1h: 23.85 ETH

- Unique traders 5m: 20

- Top 20 holder pct: 30.9%

- Time since launch: 18h

– VEXA

- Price: 0.00000196 ETH

- 5m change: +3.3%

- 1h change: +10.5%

- Net flow 5m: +1.85 ETH

- Volume 5m: 5.61 ETH

- Volume 1h: 28.27 ETH

- Unique traders 5m: 18

- Top 20 holder pct: 28.2%

- Time since launch: 32h

– MORI

- Price: 0.00000211 ETH

- 5m change: +4.2%

- 1h change: +12.3%

- Net flow 5m: +3.03 ETH

- Volume 5m: 5.26 ETH



- Volume 1h: 24.16 ETH
- Unique traders 5m: 20
- Top 20 holder pct: 31.1%
- Time since launch: 18h
- LUMA
- Price: 0.00000166 ETH
- 5m change: +2.2%
- 1h change: +6.5%
- Net flow 5m: +0.76 ETH
- Volume 5m: 3.69 ETH
- Volume 1h: 15.73 ETH
- Unique traders 5m: 11
- Top 20 holder pct: 66.8%
- Time since launch: 18h
- KIRO
- Price: 0.00000091 ETH
- 5m change: -4.7%
- 1h change: -9.3%
- Net flow 5m: +1.07 ETH
- Volume 5m: 4.34 ETH
- Volume 1h: 14.84 ETH
- Unique traders 5m: 14
- Top 20 holder pct: 32.9%
- Time since launch: 32h
- TAVO
- Price: 0.00000148 ETH
- 5m change: +7.3%

- 1h change: +4.5%
- Net flow 5m: +0.34 ETH
- Volume 5m: 1.50 ETH
- Volume 1h: 4.47 ETH
- Unique traders 5m: 4
- Top 20 holder pct: 58.2%
- Time since launch: 2h

---

#### ## ACTIVE STRATEGIES (CURRENT ONLY)

No active strategies.

---

#### ## ACTIVE SETTINGS

- Trading Activity (TA): 5 / 5 - How often you trade when there is fresh edge (1=very patient, 5=highly active). TA does NOT require a trade every tick.
  - Asset Risk Preference (Risk): 2 / 5 - Which tokens you consider (1=prefer least volatile available, 5=embrace high volatility). Risk determines which tokens to consider, not whether to trade at all.
  - Trade Size (Size): 3 / 5 - Maximum position sizing per trade (1=up to ~15%, 2=up to ~30%, 3=up to ~50%, 4=up to ~70%, 5=up to ~90% of available ETH). These are UPPER BOUNDS, not minimums.
  - Holding Style (Hold): 2 / 5 - Minimum hold time before considering an exit (1=about ~30 minutes minimum; 2=about ~1 hour; 3=hours; 4=many hours; 5=days).
  - Diversification (Div): 4 / 5 - Portfolio spread (1=concentrated in 1-2 tokens; 2=focused 2-3; 3=balanced 3-5; 4=spread 4-6; 5=wide 5+).
- 

#### ## PORTFOLIO CONTEXT

- ETH: Balance: 0.240000
- No current token holdings.

#### ## CONSTRAINTS

- Max Trade Amount (Percent): 100.00% of available ETH

#### ## PRICE IMPACT LIMITS (max 1500 bps)

- NERA: BUY max 100.00% of ETH
- VEXA: BUY max 100.00% of ETH
- MORI: BUY max 100.00% of ETH
- LUMA: BUY max 100.00% of ETH
- KIRO: BUY max 100.00% of ETH
- TAVO: BUY max 100.00% of ETH

Respond with the single best action for this tick: buy, sell, or observe.